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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab10 ( The Link Layer and LANs )

Due Date: January, 20 2026, (12:00 AM)

## Part 1: Manual Calculation

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### Exercise 1: Cyclic Redundancy Check (CRC)

#### Concept

CRC is used by network hardware to detect bit errors in transmitted frames. The sender computes CRC bits **R** such that the combined bit pattern (D, R) is exactly divisible by the generator polynomial **G** using modulo-2 arithmetic (XOR).

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#### Given

- **Data Packet (D)** = 101010 (6 bits)
  - **Generator Polynomial (G)** = 1001 (4 bits)
  - **CRC length (r)** =  $\text{length}(G) - 1 = 3 \text{ bits}$
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#### Step 1: Append Zeros to Data

Append 3 zeros to the data packet: 101010000

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#### Step 2: Modulo-2 Division

Perform binary (XOR) division of 101010000 by 1001.

**Resulting remainder:** R = 111

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#### Step 3: Transmitted Bit Pattern

Append the remainder to the original data: <D, R> = 101010111

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#### Step 4: Verification

Divide the transmitted pattern **101010111** by the generator **1001**. Remainder = 000

✓ The remainder is zero, so the frame is valid and error-free.

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## Step 5: Simulate an Error

Flip the last bit of the transmitted pattern:

- Original frame: 101010111
- Error frame: 101010110

Divide the erroneous frame by **1001**: Remainder  $\neq$  000

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## Analysis (CRC)

- The remainder is **non-zero**, meaning an error is detected.
  - The receiver uses this result to **discard the frame** and rely on higher-layer protocols for retransmission.
  - This is how CRC enables reliable error detection in hardware.
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## Exercise 2: CSMA/CD Exponential Backoff

### Concept

Ethernet uses CSMA/CD with binary exponential backoff to reduce repeated collisions. After the  $m$ -th collision, a node chooses a random value:

$$[ K \in \{0, 1, 2, \dots, 2^m - 1\} ]$$

The node waits  **$K \times 512$  bit times** before retransmitting.

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### Given

- Collision count:  **$m = 3$**
  - Chosen random value:  **$K = 4$**
  - Ethernet speed: **10 Mbps**
  - 1 bit time =  **$0.1 \mu\text{s}$**
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### Step 1: Collision Window

$$[ 2^3 - 1 = 7 ]$$

Possible values of  **$K$** : {0, 1, 2, 3, 4, 5, 6, 7}

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### Step 2: Wait Time in Bit Times

$$[ 4 \times 512 = 2048 \text{ \textit{bit times}} ]$$


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### Step 3: Convert to Real Time

$$[ 2048 \times 0.1 \mu s = 204.8 \mu s ]$$

**Total wait time before retransmission:** 204.8 microseconds

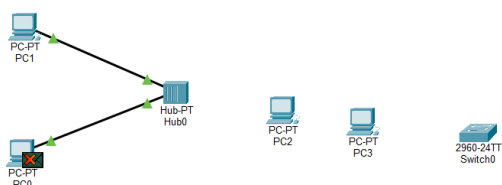
### Analysis (CSMA/CD)

- As **m increases**, the random backoff range becomes larger.
- This spreads retransmission attempts over time.
- It reduces repeated collisions and helps **relieve network congestion** when traffic is heavy.

## Activity A: Shared Media and Collisions with Hubs (Part 2)

### Topology Setup

- **Devices Used:**
  - PC0, PC1, PC2, PC3
  - Hub0
  - Switch0
- **Connections:**
  - PC0 and PC1 connected to Hub0 using Copper Straight-Through cables
  - PC2 and PC3 not connected
- **Simulation Mode:**
  - Switch to Simulation Mode (Shift + S)
  - Open "Edit Filters" and enable only **ICMP**
- **Traffic Generation:**
  - Use "Add Simple PDU" tool: - PC0 → PC1 - PC1 → PC0



| Event List |           |             |           |      |
|------------|-----------|-------------|-----------|------|
| Vis.       | Time(sec) | Last Device | At Device | Type |
| ...        | 0.000     | --          | PC1       | ICMP |
| ...        | 0.000     | --          | PC0       | ICMP |
| ...        | 0.002     | Hub0        | PC0       | ARP  |
| ...        | 0.002     | Hub0        | PC1       | ARP  |
| ...        | 2.001     | --          | PC1       | ICMP |
| ...        | 2.003     | --          | PC0       | ICMP |

Reset Simulation ☒ Constant Delay

Play Controls

Captures: 1013.11

- **Observation:**
  - Step through simulation using "Play" or "Forward"
  - Watch for collision icon at Hub0 (fire or red X)

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## Event List Summary

| Time (sec) | Last Device | At Device | Packet Type |
|------------|-------------|-----------|-------------|
| 0.000      | —           | PC1       | ICMP        |
| 0.000      | —           | PC0       | ICMP        |
| 0.002      | Hub0        | PC0       | ARP         |
| 0.002      | Hub0        | PC1       | ARP         |
| 2.001      | —           | PC1       | ICMP        |
| 2.003      | —           | PC0       | ICMP        |

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## Analysis Question

Why does a collision occur when using a Hub?

- **Layer 1 Behavior (Hub):**
  - Operates at the Physical Layer
  - Does not inspect MAC addresses
  - Repeats incoming signals to all ports
  - Creates a **shared collision domain**
  - When two devices transmit simultaneously, signals overlap → **collision**
- **Layer 2 Behavior (Switch):**
  - Operates at the Data Link Layer
  - Learns MAC addresses and forwards frames only to the correct port
  - Each port is its own collision domain
  - **No collisions** occur in full-duplex switch environments

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## Conclusion

Hubs cause collisions due to their broadcast nature at Layer 1. Switches prevent collisions by intelligently forwarding frames at Layer 2.

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## Activity B: Switch MAC Table Learning

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### Objective

Understand how switches learn device locations to create **dedicated collision domains**, unlike hubs which broadcast to all devices.

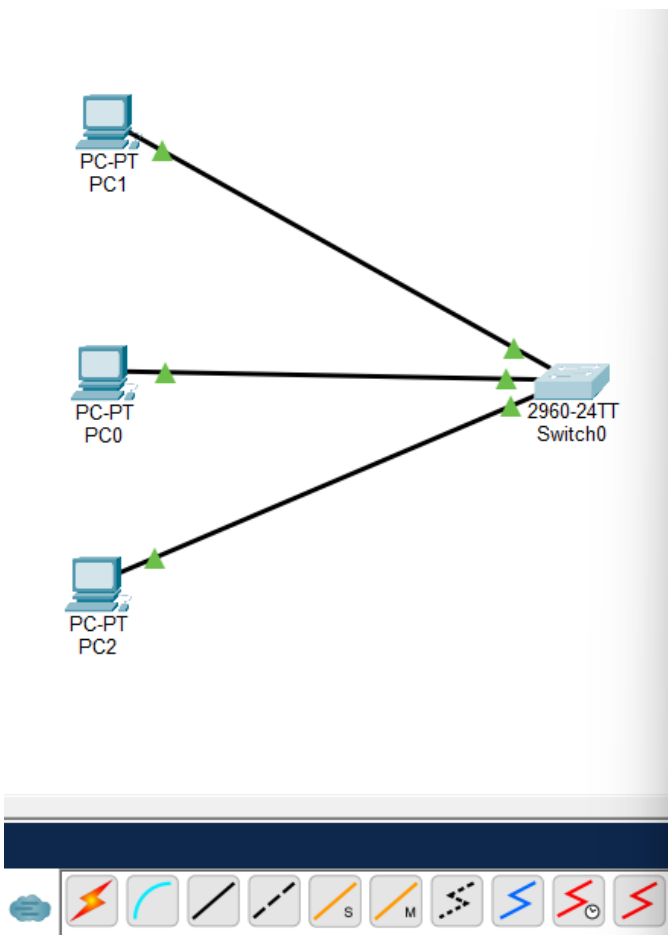
## Steps

### 1. Re-Connect

- Delete the Hub from the previous activity.
- Connect **PC0, PC1, and PC2** to the Switch.
- Wait for link lights to turn green (or press **Fast Forward Time** in Realtime mode).

### 2. Inspect Empty Table

- Open the Switch CLI.
- Run: show mac-address-table
- The table should be **empty** initially.



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Top Assembly Revision Number : A0
Version ID                   : V02
CLEI Code Number             : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version
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*    1 26    WS-C2960-24TT-L  15.0(2)SE4

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed s
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastE
%LINK-5-CHANGED: Interface FastEthernet0/2, changed s
%LINK-5-CHANGED: Interface FastEthernet0/3, changed s
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastE

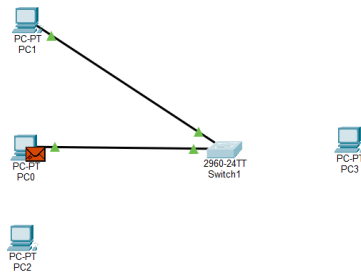
Switch>show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
Switch>
  
```

### 3. Ping

- Switch to **Simulation Mode**.
- Edit Filters: Ensure **ARP** and **ICMP** are checked.
- Use the **Add Simple PDU** tool to ping from **PC0** → **PC1**.

### 4. Step-by-Step Learning

- Click **Forward** until the frame leaves PC0 and reaches the Switch.
  - Stop and check the MAC table again: show mac-address-table
  - PC0's MAC address should now be associated with its port (e.g., **Fa0/1**).
- The switch has **learned PC0's location**.



| Vis. | Time(sec) | Last Device | At Device | Type |
|------|-----------|-------------|-----------|------|
|      | 0.000     | --          | PC0       | ICMP |
|      | 0.000     | --          | PC0       | ARP  |
|      | 0.001     | PC0         | Switch1   | ARP  |
|      | 2.003     | --          | PC0       | ICMP |

Reset Simulation ☒ Constant Delay Captured to: 1760.279 s

Play Controls ⏮ ⏪ ⏩ ⏭

## 5. Complete the Ping

- Continue forwarding until the reply returns from PC1 to PC0.
- Check the MAC table again:
- PC1's MAC address should now be learned and associated with its port.

## Analysis Question

**In Activity A (Hub), the packet was broadcast to everyone. In this activity, after the MAC table was populated, did the Switch send the ICMP reply (from PC1) to PC2? Why or why not?**

- **Answer:** No, the Switch did **not** send the ICMP reply to PC2.
- **Reason:** Once the MAC table is populated, the Switch knows which port corresponds to PC0's MAC address. It forwards the reply **only to PC0's port**.
- **Key Difference:** Unlike a Hub, which broadcasts frames to all devices, a Switch isolates traffic per port using its MAC table. This prevents collisions and unnecessary broadcasts.

## Conclusion

Switches dynamically build a MAC address table by learning source addresses from incoming frames. This allows them to forward traffic only to the correct destination port, creating **collision-free communication** and efficient use of bandwidth.

## Activity C: Switch MAC Table Learning

### Objective

Understand how switches learn MAC addresses to create dedicated collision domains and forward frames intelligently.

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## Setup Instructions

### 1. Reconnect Devices

- Delete the Hub from previous activities.
- Connect **PC0, PC1, and PC2** to **Switch0** using Copper Straight-Through cables.
- Wait for link lights to turn green or press **Fast Forward Time** in Realtime mode.

### 2. Inspect the Empty MAC Table

- Click on the Switch.
- Go to the **CLI tab**.
- Type: show mac-address-table
- The table should be **empty** initially.

### 3. Generate Traffic

- Switch to **Simulation Mode**.
- Open **Edit Filters** and ensure **ARP** and **ICMP** are checked.
- Use the **Add Simple PDU** tool to ping from **PC0** → **PC1**.

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## Step-by-Step MAC Learning

### 4. Observe Frame Movement

- Click **Forward** until the frame leaves PC0 and reaches the Switch.

### 5. Check MAC Table

- Open the Switch CLI again and run: show mac-address-table
  - You should now see **PC0's MAC address** associated with its port (e.g., Fa0/1).

### 6. Complete the Ping

- Continue clicking **Forward** until the ICMP reply returns from PC1 to PC0.

### 7. Final MAC Table Check

- Run: show mac-address-table
- You should now see **PC1's MAC address** added to the table.

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## Analysis Question

**In Activity A (Hub), the packet was broadcast to everyone. In this activity, after the MAC table was populated, did the Switch send the ICMP reply (from PC1) to PC2? Why or why not?**

- **Answer:** No, the Switch did **not** send the ICMP reply to PC2.
  - **Reason:** The Switch had already learned PC0's MAC address and its associated port. It forwarded the reply **only to PC0**, not to PC2.
  - **Conclusion:** Unlike a Hub, a Switch uses its MAC table to forward frames only to the correct destination, preventing unnecessary broadcasts and collisions.
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## Summary

Switches dynamically learn MAC addresses from incoming frames and build a table to forward traffic efficiently. This isolates communication between devices and eliminates collisions, making switches ideal for modern LAN environments.